

Kiet Vu

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EDUCATION

Purdue University

B.Sc. in Computer Engineering

West Lafayette, IN

Expected Dec. 2028

- **GPA:** 3.93/4.0
- **Expected Coursework:** ASIC Design Laboratory, Microprocessor Systems and Interfacing, Computer Design and Prototyping, Semiconductor Devices, Advanced C Programming

PROJECTS

GPU Microarchitecture Research | *Accel-Sim, CUDA, Nsight Tools, PTX/SASS*

July 2026 – Aug. 2026

- Restored PTX-mode simulation on all CUDA 11+ toolchains by fixing an Accel-Sim/GPGPU-Sim parser bug, and added an L2 write-hit counter comparable to NVIDIA's profiler; **both merged upstream**
- Root-caused a SIGSEGV in the FP64 pipeline model (out-of-bounds index when a functional unit's initiation interval exceeds its latency) and fixed a correlation bug that mis-attributed local-memory traffic, cutting L2 write error on one benchmark from **-94.0%** to **+2.6%**
- Investigated an unmodeled **4x over-count** in L1-to-L2 write traffic; built a microbenchmark to confirm the merging mechanism; prototyped a merge buffer, reducing mean L2 write error from **19.48%** to **2.40%** and L2 write-hit error from **19.10%** to **2.33%** across 32 benchmarks, with 19 benchmarks bit-identical and worst collateral change **-0.2%**

32-bit RISC-V Pipelined Processor | *SystemVerilog, Computer Architecture*

Feb. 2026 – Mar. 2026

- Architected a 5-stage pipelined RV32I processor using Harvard memory architecture with a Hazard Control Unit; supported the complete base integer instruction set, achieving **100% architectural compliance** via the RISCOF framework by co-simulating against Spike reference model
- Synthesized the core targeting the Sky130 node via LibreLane to generate a verified GDSII layout; optimized PPA trade-offs using high-speed standard cells to achieve a peak frequency of **95.2 MHz (0.551 mm²)**, alongside a high-density variant reaching **58.8 MHz (0.427 mm²)**, both with clean STA and LVS signoffs

KV-PIMple: PIM-aware KV-allocation for LLM Decode | *HBM, PIM, AI System*

Sep. 2026

- Discovered the DRAM row alignment that paged KV-cache allocation destroys for processing-in-memory (PIM) attention, raising simulated all-bank row-hit rate **from 0.77 to 0.96**, and modeled effective PIM bandwidth **2.3×** at vLLM's standard 16-token block by designing an allocator that fills one geometry-derived region
- Built a simulator of continuous-batching LLM decode, porting vLLM v0.2.7's block allocator and covering four workloads: preempt-by-recompute, speculative decoding and MQA/GQA/MHA shapes; Verified with 85 tests including five validation gates (best/worst-case anchors, monotonicity, determinism, block conservation)
- Established the model's credibility against two external cycle-level DRAM simulators by replaying its own block tables through Ramulator 2 and AttAcc's all-bank PIM extension: placement agreed to **0.0023 mean row-hit** across 9 samples on Ramulator, while timing is **1.4×** more on AttAcc due to differences in policy assumption

EXPERIENCE

Purdue System-on-Chip Extension Technologies (SoCET) GPU Team

Aug. 2026 – Present

Undergraduate Researcher

West Lafayette, IN

- Architected the Ethernet RX datapath for a fixed-function GPU on DE2-115; derived specs from IEEE 802.3, produced RTL block diagrams and state machines for frame detection, CRC-32 validation, address filtering, and CDC buffering
- Designed a custom host-to-GPU command protocol over raw Ethernet with register-write and vertex-stream

Purdue EPICS Harnessing Data Revolution

Aug. 2025 – May 2026

Undergraduate Researcher

West Lafayette, IN

- Refactored multiple legacy data source modules to standardized abstract base class architecture, implementing consistent data pipeline, improving maintainability and code reusability
- Eliminating NULL overwrites and integrating 1,800+ historical records into SQLite; centralized configuration consolidating 59 SQL field mappings, 1,000+ station IDs, and 7 database schemas across 7 data sources for 53+ monitoring locations

EXTRACURRICULAR ACTIVITIES

Purdue RoboMaster | *PCB, Robotics, Engineering*

Aug. 2025 – May 2026

- Engineered custom PCBs for a rechargeable armor power system and fluorescent projectile charger, integrating TP4056 chargers, TPS63002 buck-boost converters, ESP32 monitoring, and MOSFET/Zener-based battery protection
- Engineered an audio visualizer circuit, amplifying a microphone signal by 45x with a cascaded LM358 op-amp and driving a 10-segment LED display with an LM3915 IC

TECHNICAL SKILLS

Languages: Verilog/SystemVerilog, C/C++, Tcl, CUDA, PTX/SASS, MATLAB, Python

Computer Architecture: GPU architecture, RISC-V, I/O Interface, Ethernet, AXI, AMBA, FPGA

Software: LibreLane Toolchains, Yosys, Surfer, Linux, GitHub, EasyEDA, Altium, VS Code, Nsight Tools